1 Periodic Functions and Their Graphs

Periodic functions describe quantities that repeat as their input increases. Trigonometric functions provide fundamental examples: their periodicity arises from rotation, and their graphs can be transformed to represent a wide range of oscillatory processes.

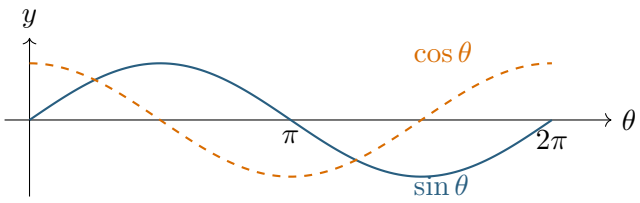
Learning objectives. Identify periods, amplitudes, midlines, and phase shifts; construct transformed sine and cosine graphs; distinguish angular frequency from frequency; and determine sinusoidal models from geometric or measured information.

1.1 Circular motion and periodicity

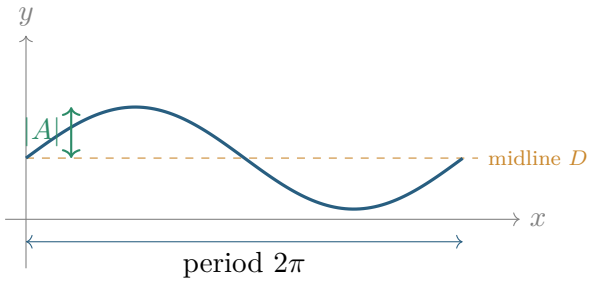
For a point traveling around the unit circle with angle θ, its vertical coordinate is $y = \sin\theta$ and its horizontal coordinate is $x = \cos\theta$. Each revolution restores the same coordinates. Hence

$$\sin(\theta + 2\pi) = \sin\theta, \qquad \cos(\theta + 2\pi) = \cos\theta.$$

A function f is *periodic* if there exists $T > 0$ such that $f(x + T) = f(x)$ for every x in its domain. When a least positive such T exists, it is the *fundamental period*. The fundamental periods of sine and cosine are both 2π. Moreover, $\sin 0 = 0$ and $\cos 0 = 1$.



1.2 Amplitude and the midline



1.3 Vertical transformations

Consider $y = A\sin x + D$. Its midline is $y = D$, its amplitude is $|A|$, and its range is $[D - |A|, D + |A|]$. A negative A reflects the wave across the midline.

Example 1.1. The function $y = 3\sin x - 2$ has amplitude 3, midline $y = -2$, maximum 1, and minimum -5.

For a sinusoid with maximum M and minimum m, the amplitude and midline are

$$|A| = \frac{M - m}{2}, \qquad D = \frac{M + m}{2}.$$

These relations determine the vertical parameters without knowing the phase or period.

Example 1.2. A periodically varying temperature has maximum 26°C and minimum 14°C. Its midline is 20°C and its amplitude is 6°C. These values do not determine the period or phase.

Try it. A sinusoidal signal ranges from -7 to 5. Determine its amplitude, midline, and range, and sketch one possible cycle.

1.4 Period and horizontal scaling

For $y = \sin(Bx)$ with $B \neq 0$, one full cycle occurs when Bx changes by 2π; the period is $T = 2\pi/|B|$.

Example 1.3. The graph of $y = \cos(4x)$ repeats every $\pi/2$ units of input.

When x denotes time measured in seconds, the quantity $|B|$ is angular frequency in radians per second, and ordinary frequency is $f = 1/T = |B|/(2\pi)$ cycles per second.

A phase change of 2π radians corresponds to one cycle. If the argument of a sinusoidal function is $\omega t + \phi$, then its period is $T = 2\pi/|\omega|$, and its frequency is

$$f = \frac{1}{T} = \frac{|\omega|}{2\pi}.$$

Frequency is measured in cycles per unit time, while angular frequency is measured in radians per unit time. These units describe the same repetition rate in different ways.

Example 1.4. A mechanism rotates at 90 revolutions per minute. Its frequency is 1.5 cycles per second, and its angular speed is $2\pi(1.5) = 3\pi$ radians per second. Its period is 2/3 second.

1.5 Phase shift

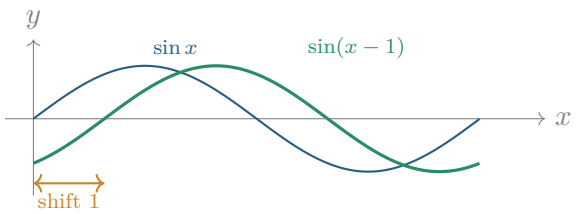
A useful general form is

$$\boxed{y = A\sin(B(x - C)) + D}.$$

Here C is the horizontal shift, D the midline, $|A|$ the amplitude, and $2\pi/|B|$ the period. We should distinguish $B(x - C)$ from $Bx - C$: factoring first prevents an incorrect phase shift.

Example 1.5. For $y = 2\sin(3x - \pi) + 1 = 2\sin(3(x - \pi/3)) + 1$, the period is $2\pi/3$, the shift is $\pi/3$ to the right, the amplitude is 2, and the midline is 1.

The parameterization assumes $B \neq 0$. In factored form, the phase shift is C. In the equivalent form $A\sin(Bx + \phi) + D$, the same shift is $C = -\phi/B$. The sign of B may be taken positive in a standard sine model by modifying the sign of A.



1.6 Five landmark points

Rather than sketching a sinusoid by guesswork, divide one period into four equal intervals. For $A > 0$, sine begins on the midline, rises to a maximum after one quarter-period, returns to the midline, reaches a minimum, then returns to the midline. A negative amplitude reverses the rises and falls.

Try it. Find five landmark points of $y = -2\cos(\pi(x-1)) + 3$, showing one entire period.

For $y = A\sin(B(x-C)) + D$ with $B > 0$, write $T = 2\pi/B$. One cycle is determined by the following five points:

x	C	$C+T/4$	$C+T/2$	$C+3T/4$	$C+T$
y	D	$D+A$	D	$D-A$	D

For cosine, the corresponding outputs are $D+A, D, D-A, D, D+A$. These points determine the positions of maxima, minima, and midline crossings; the smooth sinusoidal curve passes through them.

Example 1.6. For $y = -2\cos(\pi(x-1)) + 3$, the period is $T = 2$. Beginning at $x = 1$, the five landmark points are

$$(1,1), \quad (3/2,3), \quad (2,5), \quad (5/2,3), \quad (3,1).$$

The negative coefficient places the minimum at the start of the cycle.

1.7 Models of repeating motion

A Ferris wheel has radius 9 m, its center is 11 m above ground, and it makes one revolution every 30 s. If a passenger begins at the rightmost position and travels counterclockwise, their height is

$$h(t) = 11 + 9\sin\left(\frac{2\pi}{30}t\right).$$

Its amplitude is 9 m, midline is 11 m, and period is 30 s. A sinusoidal model can describe a wheel, a spring, a tide, or a musical tone, provided its parameters actually correspond to the chosen physical quantities.

1.8 Constructing a model from measurements

Suppose a buoy reaches a maximum height of 5 meters and a minimum height of 1 meter. Successive maxima are separated by 8 seconds. Taking $t = 0$ at a maximum yields the model

$$h(t) = 3 + 2\cos\left(\frac{\pi t}{4}\right).$$

The midline is 3 meters, the amplitude is 2 meters, and the angular frequency is $\pi/4$ radians per second. If the time origin were chosen at an upward midline crossing, a sine model would be more convenient. The physical motion is unchanged by this choice of representation.

Try it. A tide measurement varies between 0.8 m and 3.6 m and repeats every 12 hours. Write a cosine model assuming that the maximum occurs at $t = 2$ hours. State the meaning and units of every parameter.

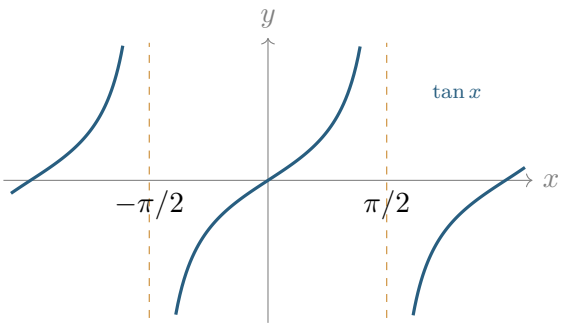
1.9 Tangent and reciprocal graphs

The tangent function repeats after π, not 2π, because both sine and cosine change sign after a half turn and their ratio remains the same:

$$\tan(x+\pi) = \tan x.$$

Tangent has vertical asymptotes at $x = \pi/2 + k\pi$. The secant and cosecant graphs arise as reciprocals of cosine and sine; their breaks occur wherever the respective denominator is zero.

Because $\tan x = \sin x/\cos x$, its zeros occur at $x = k\pi$ and its vertical asymptotes at $x = \pi/2 + k\pi$. Similarly, $\cot x = \cos x/\sin x$ is undefined at integer multiples of π. The graphs of secant and cosecant lie outside the strip $-1 < y < 1$ because their absolute values are at least 1 where defined.



1.10 Exercises

1. Find the amplitude, midline, period, and range of $y = -4\sin(2x) + 1$.
2. Rewrite $y = 3\cos(5x - \pi) + 2$ in shifted form and identify the phase shift.
3. Give five landmark points for $y = 2\sin(x - \pi/2)$.
4. Explain why $y = \sin(2x)$ completes two periods on $[0, 2\pi]$.
5. A wheel of radius 6 m rotates once every 24 seconds. Its center is 8 m above the ground. Model the rider's height when starting at the rightmost point.
6. Identify the vertical asymptotes of $y = \tan(2x)$ on $[0, 2\pi)$.
7. Sketch one period of $y = 2\sec x$ and indicate all discontinuities.
8. Explain how you can determine amplitude and midline directly from observed maximum and minimum values.
9. Determine a sinusoidal model with maximum 8, minimum -2, period 10, and maximum at $x = 3$.
10. Explain why $A\sin(B(x-C))+D$ is constant when $A = 0$, and why the usual fundamental-period formula does not apply.
11. A motor oscillates at 2.5 cycles per second. Determine its angular frequency and period.
12. Give the domain of $y = \tan(3x)$ and identify two consecutive vertical asymptotes.
13. Determine the amplitude and midline from a sinusoid with maximum 12.4 and minimum 7.6.